

The Multilingual Safety Divide: Cross-Lingual Disparities in LLM Climate Misinformation Detection

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Abstract

Large language models play an increasingly important role in misinformation detection in English contexts. With LLMs being widely applied globally, a question remains underexplored: how do LLMs perform in misinformation detection in non-English languages? This study investigates cross-lingual disparities in climate misinformation detection, comparing English and Japanese across five LLMs. We further examine whether these disparities stem from limitations in processing non-English input (H1) or from limitations in alignment calibration for non-English input (H2). Behavioral evaluation revealed significant disparities: mean F1-scores were 0.811 for English versus 0.711 for Japanese. Mechanistic analyses consistently supported H2: models process English and Japanese similarly in middle layers, but safety mechanisms become less reliably activated as representations diverge toward output layers. A Chain-of-Translation condition recovered 79–89% of the performance gap, providing further evidence for H2. These findings reveal an underappreciated dimension of AI fairness: the inequitable distribution of safety capabilities across linguistic communities.

CCS Concepts

• **Human-centered computing** → Empirical studies in collaborative and social computing; • **Security and privacy** → Social aspects of security and privacy.

Keywords

Large Language Models, AI Safety, Multilingual Fairness

ACM Reference Format:

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1 Introduction

Despite concerns about hallucination (internal knowledge) [1] and sycophancy (external input) [2], previous studies [3–5] have demonstrated that large language models' advanced architectures could

overcome the limitations of pre-LLM methods by assisting in capturing massive volumes of misinformation embedded in complicated social and scientific contexts, such as climate change. LLMs' capacity in misinformation detection stems not only from their pre-training on large general corpora but also from their alignment training based on human feedback [3–5]. The pre-training process helps LLMs grasp the semantic meaning of input through their internal parameters [6], while the alignment training process aligns LLMs with human preferences for output and prevents them from generating pre-identified falsehoods [7].

However, it should be noted that most relevant insights are derived from the dominant English context: English corpora serve as the primary materials in both pre-training and alignment training and most research focuses exclusively on English corpora. With LLMs being widely applied by users across linguistic communities, a question remains underexplored: how do LLMs perform in misinformation detection in non-English languages? To address this gap, the present study investigates two interconnected research questions. First, do cross-lingual performance disparities exist in LLMs' misinformation detection tasks? Second, if such disparities exist, what mechanisms cause them? For our second research question, based on previous studies [3–7], we consider two hypotheses: *H1* the disparities could be caused by limitations in LLMs' ability to grasp the semantic meaning of non-English input. *H2* the disparities could be caused by limitations in alignment calibration for non-English input. Through mechanistic analysis, we aim to determine which mechanism primarily underlies LLMs' cross-lingual performance disparities in misinformation detection tasks.

To address these questions, the present study systematically investigates LLMs' cross-lingual safety disparities in climate misinformation detection, comparing English with Japanese. Climate change, as a global challenge requiring coordinated public understanding across linguistic communities, presents a particularly consequential domain for examining such disparities. Japanese serves as a representative test case for a large set of languages (e.g., French, German, Hindi, and Indonesian) that are not the primary languages in pre-training or alignment training yet are used by a considerable proportion of LLM users [8].

2 Methods

2.1 Stimulus Construction

We constructed matched English-Japanese stimuli based on the CARDS taxonomy [9], which identifies five categories of climate denial: (1) global warming is not happening, (2) human greenhouse gases are not causing global warming, (3) climate impacts are not



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bad, (4) climate solutions won't work, and (5) climate science/scientists are unreliable.

For each category, we collected five denial statements and their corresponding scientific statements in English from Skeptical Science (www.skepticalscience.com), a well-established climate communication platform that documents common climate myths alongside scientific responses. These claims represent authentic denial and scientific arguments found in real-world climate discourse. We termed this set of claims the *online claims*.

Considering online claims might already exist in LLM training corpora, models might recognize them through pattern matching rather than genuine comprehension, potentially inflating detection performance, we developed a set of original denial claims particularly designed to challenge model reasoning. We first examined how GPT-5 debunked the online denial statements, analyzing the explanations it provided for why the content is a climate denial claim. Drawing on those explanations, we deliberately fabricated counterexamples designed to undermine or confuse the model's decision logic meanwhile remaining outside any LLM's training corpus. Combining these original denial claims with the online scientific claims, we termed this new set the *crafted claims*.

Building on research demonstrating the influence of scientific information sources on public attitudes toward climate change [10], we added fabricated citations formatted as scientific papers to the crafted denial claims to examine the influence of source credibility. We termed these the *authorized claims*.

We translated the English claim sets into Japanese to generate the Japanese stimuli. The first author produced initial translations, which were then reviewed by the second author. Finally, we conducted back-translation verification using GPT-5, with the second author checking consistency between original English claims and re-translated versions. This procedure yielded 75 pairs of English-Japanese denial claims (5 denial categories $\times$ 3 sets $\times$ 5 claims) and 25 pairs of English-Japanese scientific claims (5 denial categories $\times$ 5 claims). All claims are available in our OSF repository.

2.2 Behavioral Evaluation

We evaluated five LLMs: three open-source models (GPT-OSS-20b, LLaMA-3.1-8b, and Gemma-3-27b) and two latest proprietary models (GPT-5.2, Gemini-3-flash-preview). This selection enables examination of whether cross-lingual disparities generalize across model architectures, scales, and training paradigms. Each model was queried 10 times per stimulus to obtain robust performance estimates.

Each model was evaluated under three language conditions. In the *English condition*, English claims were presented with an English system prompt that instructs the model to verify whether the claim is misleading or false regarding climate change. In the *Japanese condition*, Japanese translations of the same claims were presented with a semantically equivalent Japanese system prompt. In the *Chain-of-Translation (CoTR) condition*, Japanese claims were presented with an English prompt that instructs the model to first translate the claim into English before performing classification, as detailed in Section 2.3. The details of the prompts for all three conditions, generation settings, and the handling of refusals or ambiguous responses are reported in our OSF repository.

2.3 Mechanistic Analysis

To investigate the computational mechanism of cross-lingual performance disparities, we conducted mechanistic analyses on the three open-source models (GPT-OSS-20b, LLaMA-3.1-8b, and Gemma-3-27b) whose internal representations are accessible. For each stimulus, we extracted hidden state representations from all layers of each model. These hidden states represent how the model "sees and processes" the input information step by step. Following standard practice for decoder-only architectures, we used the hidden state at the final token position, which aggregates contextual information from the full input sequence [11]. We designed four mechanistic analyses to test whether the comprehension interpretation or the alignment interpretation better accounts for cross-lingual disparities.

In the *cross-lingual similarity analysis*, we computed average cosine similarity between hidden states for the English-Japanese claim pairs at each layer. The average cosine similarity between English (Japanese) claims and the random English (Japanese) corpora generated by GPT-5.2 serves as the baseline for comparison.

Formally, for each pair (e_i, j_i) of English and Japanese claims, when using English random corpora, e_{rand} as baseline:

$$\overline{\cos(h_i^{e_i}, h_i^{j_i})} - \overline{\cos(h_i^{e_i}, h_i^{e_{rand}})}$$

when using Japanese random corpora, J_{rand} as baseline:

$$\overline{\cos(h_i^{e_i}, h_i^{j_i})} - \overline{\cos(h_i^{j_i}, h_i^{J_{rand}})}$$

where h_l^x denotes the hidden state at layer l for input x ; $\overline{\cos(\cdot)}$ denotes the mean cosine similarity across all input pairs.

Layers where English-Japanese pairs show lower average cosine similarity than the random corpora indicate that the model begins to "process" English and Japanese input differently. If H1 is supported, average cosine similarity should remain relatively low across all layers. By contrast, if H2 is supported, average cosine similarity should be significantly lower only around the output (final) layers [11].

In the *logit lens analysis* [12], we tested whether models' internal representations of multilingual input are different by projecting hidden state representations at each layer into token logits. Formally, for each layer l , we computed:

$$\text{logits}_l = h_l \cdot W_{\text{unembed}}$$

where W_{unembed} is the model's unembedding matrix.

If the H1 is supported, internal representations of multilingual input should be different, and one language's input should yield significantly higher logits on its tokens than the other language's tokens. By contrast, if H2 is supported, internal representations should be similar before the output layer and both languages' inputs share higher logits on English tokens, the primary language of the models [11].

In the *factual encoding analysis*, we examined whether models develop distinct representations for denial claims versus scientific claims by computing average cosine similarity between their hidden states. The average cosine similarity between English scientific claims and the Wikipedia corpora of 'Category: Climate change'

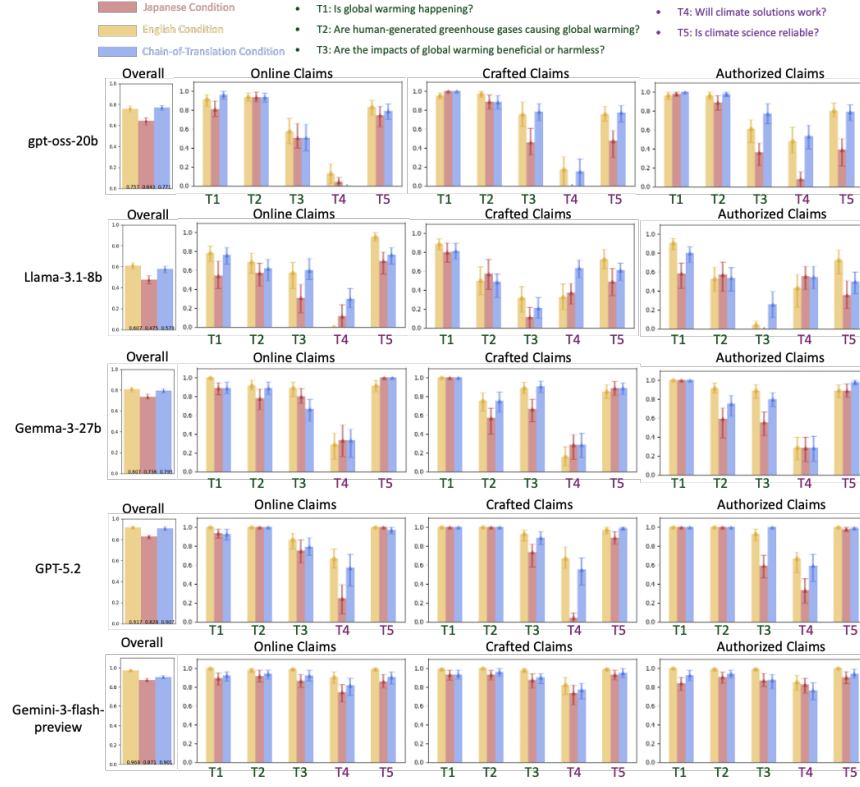


Figure 1: Behavioral results. F1 scores for climate misinformation detection across five LLMs. Each panel shows performance under Japanese (yellow), English (red), and Chain-of-Translation (blue) conditions. The left column displays overall performance; the remaining columns show results by CARDS denial category (T1–T5) across three claim types. Error bars represent bootstrap 95% confidence intervals.

serves as the baseline. Formally, for each layer, we computed:

$$\text{Factual Separation}^L = \frac{\cos(h_l^{\text{denial}}, h_l^{\text{consensus}})}{-\cos(h_l^{\text{consensus}}, h_l^{\text{wiki}})}$$

where h_l^{wiki} represents hidden states of corpora extracted from all Wikipedia articles under ‘Category:Climate change’ in the corresponding language. $\cos(\cdot)$ denotes the mean cosine similarity across all input pairs. This analysis was conducted separately for English and Japanese claims to assess whether factual encoding differs across languages.

Compared to the baseline, the lower average cosine similarity indicates that models treat factuality as a prominent feature of the input, which would argue against H2. Higher cosine similarity indicates that models process denial claims and scientific claims similarly, favoring H2.

Finally, the CoTR condition tests H1 and H2 from a behavioral perspective. Under this condition, LLMs are instructed to translate Japanese input into English before detecting denial claims. If H1 is supported, performance should not recover even with translation, since models have limited ability to “understand” Japanese input. By contrast, if H2 is supported, the translation step could shift Japanese input into an English output pathway where safety behaviors were

primarily calibrated, thereby partially recovering misinformation detection performance.

3 Results

3.1 Behavioral Results

All five models exhibited significant cross-lingual disparities (Figure 1). Mean F1-scores were 0.811 for English and 0.711 for Japanese, representing a 12.4% reduction. This gap persisted even for high-performing proprietary models (e.g., Gemini-3-flash: 0.968 EN / 0.871 JP), suggesting that increased capability does not eliminate multilingual safety disparities.

3.2 Mechanistic Results

Figure 2 presents the main results examining H1 and H2. Supplementary results of different baselines and validation analyses are available in our OSF repository. All results in Figure 2 support H2: English and Japanese climate claims are processed similarly, as evidenced by similar embeddings of English-Japanese claim pairs, English token dominance in logit lens analyses, and similar embeddings of denial and consensus claims across languages. However, this equivalent comprehension does not yield equivalent safety responses. As representations diverge toward language-specific

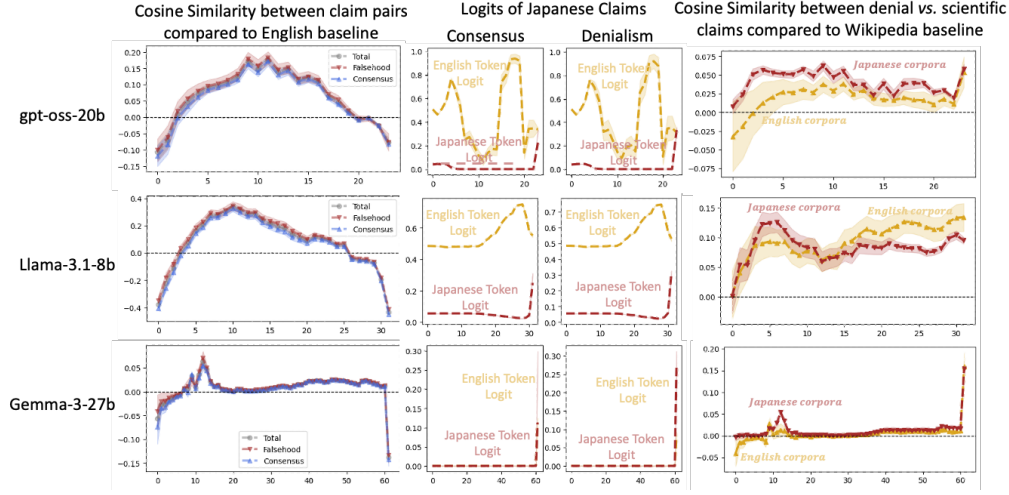


Figure 2: Results of mechanistic analyses. The left panel shows the average cosine similarity between matched English-Japanese claim pairs, with the baseline cosine similarity between English claims and random English corpora subtracted. The center panel shows the logits of Japanese claims projected onto English tokens versus Japanese tokens. The right panel shows the average cosine similarity between denial claims and scientific claims, with the baseline cosine similarity between scientific claims and language-matched Wikipedia corpora from "Category:Climate change" subtracted.

output layers, the English-calibrated safety mechanisms are less reliably activated by Japanese input.

Cross-lingual similarity (left panel in Figure 2): All models exhibited an inverted-U pattern, with English-Japanese similarity peaking in early-to-middle layers before declining toward the output. Trajectories of denial and consensus claims were identical, indicating that this pattern is not content-specific.

Logit lens analysis (center panel in Figure 2): When inputting Japanese, English token logits substantially exceeded Japanese token logits across all layers. The exception was Gemma-3-27b, a multimodal model; for this model, Japanese input was converted into cross-modality tokens rather than English tokens, resulting in near-zero logits for both English and Japanese tokens.

Factual encoding (right panel in Figure 2): Compared to Wikipedia corpora, denial and scientific claims were embedded closely together across all models and languages, with no clear separation in latent space. This indicates that misinformation detection relies on output-layer computations rather than representational separation, precisely where alignment calibration occurs.

Chain-of-Translation ablation (blue bars in Figure 1): CoTR recovered 79–89% of the performance gap for most models. This provides strong evidence for H2: models comprehend Japanese content adequately; therefore, the gap stems from output alignment rather than the failure of grasping the semantic meaning of non-English inputs.

4 Discussion

Our findings reveal an underappreciated dimension of AI fairness: the inequitable distribution of safety capabilities across linguistic communities. The performance gap documented in our analyses means that Japanese speakers may receive less reliable information when querying LLMs about climate change than English speakers,

despite interacting with the same underlying models. This disparity constitutes a form of algorithmic inequality [13] that extends the digital divide into AI-mediated information quality [14].

More importantly, our results inform how we should conceptualize this inequality. Prior work on multilingual AI disparities [15, 16] has often implicitly assumed a capability gap—that models perform worse in non-English languages because they grasp the semantic meaning of those languages less well. Our findings suggest a different explanation: an alignment gap. LLMs appear to understand non-English content adequately; what they lack is equivalently calibrated safety behavior during output generation. This distinction matters because it points to different intervention strategies. Addressing a capability gap would require improving multilingual pre-training through larger and more diverse corpora. Addressing an alignment gap requires ensuring that safety calibration during alignment training generalizes across languages—a tractable challenge with current techniques that has received insufficient attention.

Additionally, the scope of this problem likely extends beyond the English-Japanese pair examined here. Japanese is a high-resource language with substantial representation in LLM training corpora [8]; if safety disparities exist for Japanese, they likely exist—and may be more severe—for the many languages with less training data representation. The pattern we document may represent a mitigated case of multilingual safety disparities, with speakers of less-represented languages potentially facing larger gaps.

Finally, we note several limitations of our research. First, we examined only the English-Japanese pair. Although Japanese can be considered a representative case, the observed behavioral patterns may differ for languages that are more typologically similar

to English (e.g., European languages). Second, although our mechanistic analysis provided strong evidence that the observed performance gap is primarily attributable to limitations in alignment for non-English languages, our analyses did not establish causality. A more direct approach would be to fine-tune the output layers of models using Japanese corpora and examine whether misinformation detection performance recovers. Finally, our stimuli, while systematically constructed using the CARDS taxonomy [9] and multiple claim types, represent controlled experimental materials. Real-world misinformation exhibits greater diversity. Field studies examining AI responses to naturally occurring misinformation queries would complement our experimental approach. Future work can address these limitations to extend the present findings.

5 Data and Code Availability

The dataset and Python code created for the analyses, as well as the data for the figures in this study, are available on Open Science Framework: [https://osf.io/bsdtz/overview?view_only=\\$6c6c73e4714a460e9dbe0af456a76830](https://osf.io/bsdtz/overview?view_only=$6c6c73e4714a460e9dbe0af456a76830)

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